1 Site Displacement and Geophysical Corrections

The receiver coordinate that enters the line-of-sight construction and the linearised geometric range of the Foundations document is not the instantaneous position of the crust. It is the conventional tide-free station position, to which the IGS/ITRF products are referenced, plus a sum of *a priori* geophysical corrections. Each correction is a modelled 3D displacement $\boldsymbol{d}(t)$ of the antenna reference point. The ARP is the position state, which the engine obtains from the RINEX monument and the header eccentricity (the marker/ARP section of the Coordinate Frames document). None of the corrections is estimated. All are computed from the a priori position (good to centimetres), the Sun and Moon ephemerides, and the Earth-orientation parameters. They are then projected onto the line of sight and added to the modelled range. This document derives the models as the IERS 2010 conventions [2] intend them, so it can be used to settle questions of sign and phase in the implementation.

Write the corrected receiver position as

$$\boldsymbol{r} = \boldsymbol{r}_0 + \boldsymbol{d}_{\mathrm{SET}} + \boldsymbol{d}_{\mathrm{OTL}} + \boldsymbol{d}_{\mathrm{PT}} + \boldsymbol{d}_{\mathrm{OPT}} + \boldsymbol{d}_{\mathrm{APL}}, \tag{1}$$

with $\boldsymbol{r}_0$ the conventional a priori position and each $\boldsymbol{d}$ an ECEF vector of millimetres to decimetres. The five terms are:

- the solid Earth tide $\boldsymbol{d}_{\mathrm{SET}}$, the largest, up to $\sim$30 cm radially;
- ocean tidal loading $\boldsymbol{d}_{\mathrm{OTL}}$, centimetres near coasts;
- the solid (rotational-deformation) pole tide $\boldsymbol{d}_{\mathrm{PT}}$, below a centimetre to a centimetre, secular plus Chandler;
- the ocean pole tide $\boldsymbol{d}_{\mathrm{OPT}}$, below a millimetre to $\sim$2 mm;
- atmospheric pressure loading $\boldsymbol{d}_{\mathrm{APL}}$, 2–20 mm.

Only the sun-synchronous S_1/S_2 thermal-tide part of the atmospheric loading is modelled. The non-tidal (synoptic) part is a known gap (Section 1.6). All five are computed in `src/core/corrections/GnssCorrections.cpp` and `src/core/corrections/OceanTideModel.cpp`. The LOS projection and the summation into the modelled range are in `src/core/model/ModelGeometry.cpp` and `src/core/model/PppModel.cpp`. Each term has a config flag (`enable_solid_earth_tide`, `enable_ocean_tide_loading`, `enable_pole_tide`, `enable_ocean_pole_tide`, `enable_atmospheric_loading`, all under `model_corrections`). With a flag off, that term is a zero vector. The loader defaults are the solid Earth tide, ocean tidal loading and the pole tide on, and the ocean pole tide and the atmospheric loading off; ocean tidal loading without a coefficient source is a zero vector (Section 1.3.2). The real-data reference configuration `config/golden/golden_field_refound.json` arms the solid tide, the pole tide and ocean tidal loading with its three extensions (Section 1.3); the simulated-network reference configuration `config/golden/golden_sim_refound.json` leaves ocean loading off, because the simulated data carry none.

1.1 Sign convention and projection into the range

Every displacement is projected onto the unit line of sight before it enters the modelled range. Let $\boldsymbol{e} = (\boldsymbol{r}^i - \boldsymbol{r})/\|\boldsymbol{r}^i - \boldsymbol{r}\|$ be the receiver-to-satellite unit vector (the convention of the Foundations LOS equation; see `ModelGeometry.cpp` where `los = (sat_pos -rx_pos)/range`). A site displacement $\boldsymbol{d}$ changes the geometric range by its component along the line of sight. The range shortens when the antenna moves *toward* the satellite. The scalar correction added to the modelled range is

$$\boxed{\delta\rho \;=\; \boldsymbol{d}\cdot(-\boldsymbol{e}) \;=\; \boldsymbol{d}\cdot\frac{\boldsymbol{r}-\boldsymbol{r}^i}{\|\boldsymbol{r}-\boldsymbol{r}^i\|},} \tag{2}$$

so a displacement toward the satellite (a positive component along $\boldsymbol{e}$) gives $\delta\rho < 0$. The ocean-loading, atmospheric-loading, and pole-tide projections use this form (`Geometry::computeSatelliteGeometry`: `otl_corr = otl_ecef.dot(-los)`, `apl_corr = apl_ecef.dot(-los)`; `PppModel::computePoleTideProjection`: `pole_tide_ecef.dot(-los)`), and the modelled range adds them in (`PppModel::computeBaseModeledValue` return sums `+ otl_corr + apl_corr + solid_tide` and the pair processors add `+ pole_tide_corr`). The solid-Earth-tide helper does its own projection with the same vector $(\boldsymbol{r}-\boldsymbol{r}^i)/\|\boldsymbol{r}-\boldsymbol{r}^i\|$ (`GnssCorrections.cpp`, `computeSolidEarthTide`: `los = (receiver_pos -sat_pos)/rho_norm`, then `disp.dot(los)`), so all the tide terms use one sign convention. The rest of this document gives each model as a 3D ECEF (or local) displacement $\boldsymbol{d}$ and leaves Equation (2) implicit.

The Sun and Moon positions $\boldsymbol{r}^{\odot}, \boldsymbol{r}^{\mathrm{m}}$ on which the body tides depend, and the ECI↔ECEF rotation and Earth-orientation parameters they require, come from `src/core/model/Planetary.cpp` and are documented in the Coordinate Frames and Time Systems document. Here $\boldsymbol{r}^{\odot}, \boldsymbol{r}^{\mathrm{m}}$ are given geocentric ECEF vectors.

1.2 Solid Earth tide (IERS 2010, Steps 1 + 2)

The solid Earth tide is the elastic deformation of the whole Earth under the lunisolar tide-generating potential. The implementation is a port of the IERS Conventions `dehanttideinel` routine [2] and has two steps. Step 1 is evaluated in the time domain from the instantaneous Sun and Moon geometry. Step 2 is a frequency-domain correction for the few constituents whose Love and Shida numbers differ from the nominal degree-2 values. **The whole Step 1+2 block is validated against reference values**: it reproduces the official IERS `dehanttideinel` test vector and the in-tree POINT reference to below a millimetre, cross-checked by `scripts/solid_tide_reference.py` (`GnssCorrections.cpp`, `computeSolidEarthTideDisplacement` and the `SolidEarthTideTest` suite). The permanent (zero-frequency) tide is *retained*, so the corrected coordinate is conventional tide-free, as in IGS/ITRF (see the routine's header comment).

1.2.1 Step 1: in-phase degree 2 and 3

For each tide-raising body $b \in \{\odot, \mathrm{m}\}$ at geocentric position $\boldsymbol{r}^b$ ($r_b = \|\boldsymbol{r}^b\|$) and the station at $\boldsymbol{r}_0$ ($r = \|\boldsymbol{r}_0\|$, geocentric latitude φ via direction cosines), the degree-2 in-phase displacement is built from the cosine of the body zenith angle $\cos Z_b = \hat{\boldsymbol{r}}_0 \cdot \hat{\boldsymbol{r}}^b$ and a body scale factor

$$\mathrm{fac}_b \;=\; \frac{M_b}{M_\oplus}\, R_e \left(\frac{R_e}{r_b}\right)^3, \tag{3}$$

with $R_e = 6378136.55$ m and mass ratios $M_\odot/M_\oplus = 332945.943062$, $M_{\mathrm{m}}/M_\oplus = 0.012300034$ (`GnssCorrections.cpp` tide-helper constants `kTideRe`/`kTideMassRatioSun`/`kTideMassRatioMoon`).

The radial and tangential weights are

$$p_2^b = 3\Big(\tfrac{h_2}{2} - l_2\Big)\cos^2 Z_b - \tfrac{h_2}{2}, \tag{4}$$

$$x_2^b = 3\,l_2 \cos Z_b, \tag{5}$$

(`GnssCorrections.cpp`, `computeSolidEarthTideDisplacement`), where h_2, l_2 are the *latitude-dependent* degree-2 Love and Shida numbers of IERS Eq. 7.2,

$$h_2 = 0.6078 - 0.0006\,(1 - \tfrac{3}{2}\cos^2\varphi), \qquad l_2 = 0.0847 + 0.0002\,(1 - \tfrac{3}{2}\cos^2\varphi) \tag{6}$$

(`GnssCorrections.cpp`, `computeSolidEarthTideDisplacement`). An analogous degree-3 term uses fixed $h_3 = 0.292$, $l_3 = 0.015$ (`kTideH3`/`kTideL3`) and $\mathrm{fac}_b^{(3)} = \mathrm{fac}_b\,(R_e/r_b)$ (same routine). The Step-1 in-phase displacement sums the radial weight along the station unit vector and the tangential weight along each body's unit vector:

$$\boxed{\boldsymbol{d}^{(1)}_{\mathrm{SET}} = \sum_{b\in\{\odot,\mathrm{m}\}} \mathrm{fac}_b\Big(x_2^b\,\hat{\boldsymbol{r}}^b + p_2^b\,\hat{\boldsymbol{r}}_0\Big) \;+\; (\text{degree-3 analogue}),} \tag{7}$$

matching the per-component accumulation in `computeSolidEarthTideDisplacement`.

1.2.2 Step 1: out-of-phase anelasticity and the L^1 term

Mantle anelasticity makes the diurnal and semidiurnal degree-2 Love numbers complex; the small imaginary parts produce out-of-phase displacements `ST1IDIU` (diurnal, $\delta h = -0.0025$, $\delta l = -0.0007$) and `ST1ISEM` (semidiurnal, $\delta h = -0.0022$, $\delta l = -0.0007$) (`GnssCorrections.cpp`, helpers `st1Diurnal`, `st1SemiD`). The latitude (L^1) dependence of the Love numbers contributes a horizontal-only correction `ST1L1` with $l^{(1)}_{\mathrm{diu}} = 0.0012$, $l^{(1)}_{\mathrm{sem}} = 0.0024$ (helper `st1L1`). These three helpers return local radial/north/east displacements rotated into ECEF and are added to Equation (7) inside `computeSolidEarthTideDisplacement`. Their trigonometric forms follow the IERS reference and are not reproduced here.

1.2.3 Step 2: frequency dependence (Doodson tables)

Step 2 corrects the constituents whose actual Love numbers differ from the nominal degree-2 value, summing harmonic terms keyed by the Doodson fundamental arguments. The astronomical arguments τ, s, h, p, N', p_s (lunar time, Moon/Sun mean longitudes, lunar perigee, negative node, solar perigee) are polynomials in Julian centuries T from J2000, with the 15°/hr Earth-rotation term entering τ through the fractional UT hour H:

$$s = 218.31664563^\circ + 481267.88194\,T - 0.0014663889\,T^2 + \ldots, \tag{8}$$

$$\tau = 15^\circ H + 280.4606184^\circ + 36000.7700536\,T + \ldots - s, \tag{9}$$

with the remaining h, p, N', p_s polynomials given to fourth order in `GnssCorrections.cpp` (`computeAngles`). For each table line the phase is $\theta = \tau + n_s s + n_h h + n_p p + n_{N'} N' + n_{p_s} p_s (+\,\lambda$ for diurnal) and the in-/out-of-phase radial and tangential amplitudes (millimetres) are applied with a latitude weight.

Diurnal band (Table 7.3a). Thirty-one lines, dominated by K_1 and P_1 (`GnssCorrections.cpp`, `st2Diurnal`). The displacement for one line is

$$\begin{aligned} \delta R &= (A_R^{\text{in}} \sin\theta + A_R^{\text{out}} \cos\theta)\, 2 \sin\varphi \cos\varphi, \\ \delta N &= (A_T^{\text{in}} \sin\theta + A_T^{\text{out}} \cos\theta)\, \cos 2\varphi, \\ \delta E &= (A_T^{\text{in}} \cos\theta + A_T^{\text{out}} \sin\theta)\, \sin\varphi, \end{aligned} \tag{10}$$

(`st2Diurnal`), summed over the table and rotated radial/north/east $\rightarrow$ ECEF. The reference amplitudes (mm) include $K_1 : (12.00, -0.80, -0.67, -0.03)$ and $P_1 : (-1.23, -0.07, 0.06, 0.01)$ in the order $(A_R^{\text{in}}, A_R^{\text{out}}, A_T^{\text{in}}, A_T^{\text{out}})$ (the `st2Diurnal` table `T`).

Long-period band (Table 7.3b). Five zonal lines, with no longitude or τ dependence and no east component (`GnssCorrections.cpp`, `st2Lon`). The radial weight is the zonal Legendre form $(3\sin^2\varphi - 1)/2$ and the north weight is $\sin 2\varphi$:

$$\delta R = (A_R^{\text{in}} \cos\theta + A_R^{\text{out}} \sin\theta)\, \frac{3\sin^2\varphi - 1}{2}, \qquad \delta N = (A_T^{\text{in}} \cos\theta + A_T^{\text{out}} \sin\theta)\, \sin 2\varphi, \tag{11}$$

(`st2Lon`). The full solid-tide displacement is $\boldsymbol{d}_{\text{SET}} = \boldsymbol{d}^{(1)}_{\text{SET}}$ + ST1IDIU + ST1ISEM + ST1L1 + STEP2DIU + STEP2LON (`computeSolidEarthTideDisplacement`).

1.3 Ocean tidal loading (OTL)

Ocean tidal loading is the elastic response of the crust to the moving mass of the ocean tides. It reaches centimetres near coasts and is computed as a harmonic sum over the eleven principal constituents M2, S2, N2, K2, K1, O1, P1, Q1, Mf, Mm, Ssa (`OceanTideModel.hpp`, `OTL_CONSTITUENT_NAMES`). No admittance interpolation to minor constituents is applied. Each constituent k contributes, on each local component $c \in \{\text{up}, \text{west}, \text{south}\}$,

$$\boxed{dr_c(t) \;=\; \sum_{k=1}^{11} f_k(t)\, A_{c,k}\, \cos\big(\chi_k(t) + u_k(t) - \phi_{c,k}\big),} \tag{12}$$

where $A_{c,k}$ is the station amplitude, $\phi_{c,k}$ the Greenwich phase lag, and $\chi_k(t)$ the astronomical argument (the eleven-constituent accumulation of `OceanTideModel::computeDisplacement`). The nodal factor f_k and nodal phase u_k are exactly 1 and 0 unless `model_corrections.otl_nodal_corrections` is set (Section 1.3.3), and the sum is then the original evaluation bit for bit. The internal accumulation is in the $(\text{up}, \text{west}, \text{south})$ frame; the conversion to ENU is $(E, N, U) = (-\text{west}, -\text{south}, \text{up})$, then ENU$\rightarrow$ECEF (same routine). A non-finite displacement, or one above 0.5 m, is replaced by zero.

1.3.1 Astronomical argument and the Doodson–Warburg phase bias

The argument χ_k is assembled from Doodson multipliers $[\,\tau, s, h, p, N', p_s\,]$ acting on the variables $[\,\text{GMST} + \pi,\ s,\ h,\ p,\ N',\ p_s\,]$, themselves derived from the Delaunay fundamental arguments $s = F + \Omega,\ h = s - D,\ p = s - l,\ N' = -\Omega$ (`OceanTideModel::computeTidalArguments`). The BLQ Greenwich phase lags and the EOT20/FES complex coefficients are both referenced to a tide argument that carries a *species-dependent Doodson–Warburg phase bias*. For the diurnal band the

bias is a quarter cycle ($\pm 90°$), because the Cartwright–Tayler–Edden tide-generating potential refers to a *sine* rather than a cosine when (degree $+ n_\tau$) is odd. The bias per constituent (cycles) is

M2	S2	N2	K2	K1	O1	P1	Q1	Mf	Mm	Ssa
0	0	0	0	+0.25	−0.25	−0.25	−0.25	0	0	0

(13)

applied as $\chi_k \mathrel{+}= 2\pi\,(\text{bias}_k)$ (the `CONSTITUENT_PHASE_BIAS_CYCLES` step of `OceanTideModel::computeTidalArguments`; the table itself in `OceanTideModel.hpp`). **Without this bias every diurnal constituent (K1/O1/P1/Q1) is wrong in phase by** 90°, and the whole diurnal OTL contribution is mismodelled. The bias must be present, with the signs of Equation (13). This settles the question of the diurnal OTL phase.

1.3.2 Coefficient sources: SIF, BLQ, and the EOT20 grid

The per-station amplitudes and phases come from the first available of four sources (`Geometry::loadData`, OTL coefficient block): a Station Information File (SIF), a station BLQ block, the global EOT20 grid, and finally zero.

The SIF is consulted when `processing.enable_station_info` or `model_corrections.otl_from_sif` is set. The second setting loads the registry for ocean loading alone (from `files.station_info_path(s)`, or `static/sif/` of the data root), without arming the a priori position and the ocean pole tide of the station-information subsystem. A prefix match of the marker is confirmed by the receiver's approximate position (the engine section of the Reference Frames document). When the SIF has an OTL block for the marker, it supplies per-station three-component FES2014b coefficients, the best of the four sources. The block is generated from BLQ files, whose rows are radial, west-positive and south-positive, and stores them under the names UP, EAST and NORTH. The model accumulates in Up/West/South (Equation (12)). Two readings of the horizontal columns exist (`SifOtlHorizontal`, `otlCoefficientsFromSif`). Under `otl_from_sif` the EAST and NORTH columns are the BLQ west and south rows and are copied unchanged, as NGS M-PAGES passes them to its HARDISP routine. Two independent checks support that reading (`scripts/otl_compare/sif_horizontal_convention.py` and `sif_vs_blq_rows.py`): the horizontal vector of a station points up the gradient of the radial depression measured from its neighbours in 98 % of 568 stations for M2, and the Onsala semidiurnal columns match the CSR4.0 west and south rows without a shift. With `enable_station_info` alone, the columns are read as true east and north. Since East = −West and South = −North, the horizontal components then keep their amplitude and shift phase by +180°. That reading reverses the horizontal vector. It is kept so that runs that armed the subsystem stay unchanged. Up is copied in both readings. A marker with no OTL block in the SIF falls through to the BLQ block and then to the grid, with a warning.

Without a SIF entry the coefficients come from a station BLQ block. Without that, they are synthesised by bilinear interpolation of the global EOT20 complex load grid, which gives the radial component only. The grid is sampled at the RINEX header *marker* position (`RinexParser::getApproxPositionXyz`), not at the ARP position state. The difference is immaterial at this grid resolution, but the two points are distinct in the code. With no source the displacement is zero.

The EOT20 netCDF packs the real and imaginary load coefficients as `int16` with a `scale_factor`/`add_offset`. The netCDF C library does *not* apply these automatically, so the code unpacks them as $v = v_{\text{int16}} \cdot \text{scale} + \text{offset}$ (`OceanTideModel::parseEot20Load`: its `getScaleOffset` lambda and the manual unpack that follows). The scale is $\approx 5.34 \times 10^{-5}$ m/LSB, and omitting it inflates the amplitudes by ∼20000×. The complex load $Z = \text{Re} + i\,\text{Im}$ yields the real signal

$h(t) = \mathrm{Re}\{Z\, e^{i\chi}\} = A\cos(\chi - \phi)$ with

$$A = \sqrt{\mathrm{Re}^2 + \mathrm{Im}^2}, \qquad \phi = \mathrm{atan2}(-\mathrm{Im},\, \mathrm{Re}) \tag{14}$$

(ϕ in BLQ format being the Greenwich phase lag in degrees; same routine). Because EOT20 orders its 17 constituents differently from the BLQ convention, the synthesis remaps the BLQ slot k to the EOT20 index via $\{4, 13, 8, 3, 2, 9, 10, 11, 6, 7, 15\}$ (the `EOT20_INDICES_FOR_BLQ` table in `OceanTideModel::parseEot20Load`).

1.3.3 Nodal corrections and the UTC Greenwich argument

Each constituent of (12) is one spectral line. The 18.6-year regression of the lunar node Ω splits a lunar line into a main line and side lines that differ from it only in the multiplier n_l of Ω, and the sum of the lines modulates the amplitude and advances the phase of the constituent. With `model_corrections.otl_nodal_corrections`, constituent k is scaled by f_k and advanced by u_k at the node longitude Ω of the epoch,

$$f_k\, e^{\,\mathrm{i}\, u_k} \;=\; 1 + \sum_{l \in \text{side lines of } k} r_{k,l}\, e^{-\mathrm{i}\, n_l\, \Omega}, \tag{15}$$

with $r_{k,l}$ the signed ratio of the equilibrium-tide amplitude of a side line to that of the main line (`OceanTideModel::nodalFactor`). The ratios are those of the Cartwright–Tayler–Edden development tabulated in the IERS HARDISP routine, and the loading admittance is taken as flat across a main line and its side lines. The series reproduces the usual closed forms (for M2, $f = 1 - 0.0373\cos\Omega$ and $u = -2.14^\circ \sin\Omega$) to 0.0005 in f and 0.1° in u for M2, N2, K1, O1, Q1 and Mf.

The same setting takes the Greenwich sidereal angle inside χ_k from the UTC Julian Date and not from the GPS-scale date, as HARDISP does. GPS time leads UTC by 18 s, which shifts a semidiurnal argument by 0.15°; UT1 − UTC (below 0.9 s) is neglected, as in HARDISP. Against HARDISP the evaluation agrees within 0.2 to 1.9 mm over six hours at four stations (`docs/refound/OTL_ocean_tide_loading_source_2026-10-16.md`).

1.3.4 The net translation of the diurnal loading

The SIF loading is computed relative to the centre of mass of the Earth system. Part of the diurnal loading moves every station by the same ECEF displacement, a translation of the origin: a uniform vector fitted to the 597 stations of the bundled IGS SIF carries 45 % (K1), 40 % (O1) and 45 % (P1) of the east, north and up variance, against 6 % for the semidiurnal species. Orbit and clock products whose solution absorbs a common network translation do not contain that part, so a user working against them does not see it in the data, and applying it moves the estimated position by minus the translation. `model_corrections.otl_remove_diurnal_translation` (with `otl_from_sif`) removes it from the coefficients of the four diurnal constituents K1, O1, P1 and Q1. Let $Z_{c,k} = A_{c,k}\, e^{-\mathrm{i}\phi_{c,k}}$ be the BLQ phasor of component c and $\boldsymbol{G}_k$ the fitted translation phasor of constituent k, one complex number per ECEF axis (`removeOtlDiurnalTranslation`; the table is generated by `scripts/otl_compare/net_translation.py`). The east, north and up components of $\boldsymbol{G}_k$ at the station are g_k^E, g_k^N, g_k^U, and

$$Z_{\mathrm{up},k} \leftarrow Z_{\mathrm{up},k} - g_k^U, \qquad Z_{\mathrm{west},k} \leftarrow Z_{\mathrm{west},k} + g_k^E, \qquad Z_{\mathrm{south},k} \leftarrow Z_{\mathrm{south},k} + g_k^N, \tag{16}$$

since east is minus west and north is minus south. The semidiurnal and long-period constituents are unchanged. The table holds for the SIF it was fitted to, and the function applies to the

coefficients that `otlCoefficientsFromSif` returns, not to a BLQ file or the grid. On user replays of CODE-based network products the scale at which the north component of the translation appears in the data is -0.11 (95 % interval -0.75 to $+0.27$), against 1.19 for the rest of the model (same document).

Armed state. The real-data reference configuration sets `enable_ocean_tide_loading`, `otl_from_sif`, `otl_nodal_corrections` and `otl_remove_diurnal_translation`, and reads the registry from `static/sif/` of the data root. Over a whole tidal day the four settings gain 0.8 mm at the test station (`docs/refound/BOF_built_off_follow_through_2026-10-28.md`).

1.4 Pole tide (rotational deformation, IERS 7.1.4)

The solid pole tide is the elastic deformation caused by the centrifugal potential of polar motion. As the rotation pole moves relative to its secular mean, the bulge it raises moves with it. The wobble is the polar-motion offset from the IERS 2018 secular mean pole. In milliarcseconds, with t_y the decimal year,

$$\bar{x}_p = 55.0 + 1.677\,(t_y - 2000), \qquad \bar{y}_p = 320.5 + 3.460\,(t_y - 2000) \tag{17}$$

(`GnssCorrections.cpp`, `computePoleTideCorrection`). The post-2018 IGS products used by the engine are referenced to this linear model. The wobble components (arcseconds) are

$$m_1 = x_p - \bar{x}_p, \qquad m_2 = -\big(y_p - \bar{y}_p\big), \tag{18}$$

with x_p, y_p the polar-motion values linearly interpolated from the ERP table at the query epoch (`computePoleTideCorrection`; the $\bar{x}_p, \bar{y}_p$ millarcsecond means are scaled to arcseconds in the difference). **The wobble must be in arcseconds, not radians.** The displacement coefficients below are in metres per arcsecond, so a wobble in radians scales the whole pole tide by $\sim 4.85 \times 10^{-6}$. The wobble comment in `computePoleTideCorrection` records this as a bug that was once present.

IERS 2010 Eq. 7.24 gives the rotational-deformation displacement in colatitude $\theta = 90^\circ - \varphi$; substituting geodetic latitude ($\sin 2\theta = \sin 2\varphi$, $\cos 2\theta = -\cos 2\varphi$, $\cos\theta = \sin\varphi$, north $= -S_\theta$) yields the radial, north, and east components (metres):

$$\boxed{\begin{aligned} \boldsymbol{d}^R_{\mathrm{PT}} &= -0.033\,\sin 2\varphi\,\big(m_1\cos\lambda + m_2\sin\lambda\big), \\ \boldsymbol{d}^N_{\mathrm{PT}} &= -0.009\,\cos 2\varphi\,\big(m_1\cos\lambda + m_2\sin\lambda\big), \\ \boldsymbol{d}^E_{\mathrm{PT}} &= +0.009\,\sin\varphi\,\big(m_1\sin\lambda - m_2\cos\lambda\big), \end{aligned}} \tag{19}$$

(`computePoleTideCorrection`). **The east factor is $\sin\varphi$ ($= \cos\theta$), not $\cos\varphi$.** A factor of $\cos\varphi$ is inconsistent with the radial and north terms, which use geodetic φ. It gives a scale error of $\sim 1.48\times$ at mid-latitude and the wrong latitude dependence, vanishing at the equator instead of the pole. This settles the question of the pole-tide east term. The local (R, N, E) vector is rotated into ECEF at the end of `computePoleTideCorrection`.

1.5 Ocean pole tide (Desai 2002, IERS 7.1.5)

Polar motion also redistributes ocean mass. The load of this ocean pole tide deforms the crust by a further amount, from below a millimetre to $\sim$2 mm. It is added to Equation (19) when the station's six Desai grid load coefficients are available (the `has_opt_coeffs_` branch of

`computePoleTideCorrection`). Following Desai (2002) [1], the same wobble m_1, m_2 drives a complex pole-tide coefficient

$$c^R = m_1\,\gamma_2^R + m_2\,\gamma_2^I, \tag{20}$$

$$c^I = m_2\,\gamma_2^R - m_1\,\gamma_2^I, \tag{21}$$

with the IERS/Desai load Love combination $\gamma_2^R = 0.6870$, $\gamma_2^I = 0.0036$ (`kOptGammaReal`/`kOptGammaImag`). Each of the six per-station coefficients u_c^R, u_c^I (north, east, radial real and imaginary parts), bilinearly interpolated from the Desai grid, gives the local displacement

$$\boxed{\boldsymbol{d}^c_{\mathrm{OPT}} = \kappa\left(c^R\,u_c^R + c^I\,u_c^I\right), \qquad c \in \{E, N, U\},} \tag{22}$$

(`GnssCorrections.cpp`, `computeOceanPoleTideDisplacement`). The scale κ combines the IERS load-deformation constants with the arcsecond-to-radian conversion, so the wobble stays in arcseconds as in Equation (18):

$$\kappa = \frac{4\pi\,G\,a\,\rho_w\,H_p}{3\,g_E}\,\frac{\pi}{648000}, \qquad H_p = \sqrt{\tfrac{8\pi}{15}}\,\frac{\Omega^2 a^4}{GM_\oplus}, \tag{23}$$

with $\rho_w = 1025\ \mathrm{kg\,m^{-3}}$, $a = 6378136.6$ m, $g_E = 9.7803278\ \mathrm{m\,s^{-2}}$, $\Omega = 7.292115 \times 10^{-5}\ \mathrm{rad\,s^{-1}}$, $G = 6.67428 \times 10^{-11}$, $GM_\oplus = 3.986004418 \times 10^{14}$ (the `kOpt*` constants; κ assembled in `oceanPoleTideKappa`). The whole ocean-pole-tide path reproduces the NGS M-PAGES implementation and the independent Python oracle `scripts/ocean_pole_tide_reference.py` to $\sim 10^{-12}$ (see the constants' header comment).

1.6 Atmospheric pressure loading (S1/S2 thermal tide)

Atmospheric pressure loading (APL) is the response of the crust to changes in surface pressure. It is ∼2–20 mm and mostly vertical. It has two parts. The S_1/S_2 tidal part is sun-synchronous and thermally driven (diurnal and semidiurnal), and is predictable from local solar time. The non-tidal (synoptic) part is driven by weather systems. The engine models the *tidal* part only; the non-tidal part is a gap (see the note below).

The thermal-tide displacement uses per-station diurnal (S_1) and semidiurnal (S_2) harmonic coefficients evaluated against the local-solar-time argument $\Theta_n = n\,(2\pi\,\mathrm{UT_{frac}} + \lambda)$, $n \in \{1, 2\}$, with $\mathrm{UT_{frac}}$ the UT fraction of day and λ the station longitude (`GnssCorrections.cpp`, `computeAtmosphericTideDisplacement`):

$$\boxed{\boldsymbol{d}^c_{\mathrm{APL}} = a^{\cos}_{c,S_1}\cos\Theta_1 + a^{\sin}_{c,S_1}\sin\Theta_1 + a^{\cos}_{c,S_2}\cos\Theta_2 + a^{\sin}_{c,S_2}\sin\Theta_2, \qquad c \in \{E, N, U\},} \tag{24}$$

formed in local ENU and rotated to ECEF, with a 1 cm sanity guard. The twelve coefficients per station are ordered $[\,a^{\cos}_E, a^{\sin}_E, a^{\cos}_N, a^{\sin}_N, a^{\cos}_U, a^{\sin}_U\,]$ for S_1 (indices 0–5) then the same six for S_2 (indices 6–11). They are bilinearly interpolated from a Ray–Ponte-style S_1/S_2 load grid (`loadAtmosphericTideFromGrid`) at the same RINEX header marker position as the OTL grid (Section 1.3.2). The per-epoch displacement is evaluated at the ARP position state. The term is gated by `model_corrections.enable_atmospheric_loading` (default `false`). It reads its grid from `files.atm_tide_grid_path`, which defaults to `<data>/static/apl/s1_s2_load_cm.dat`. That filename is PROVISIONAL and will be set to the supplied product. The coefficients are loaded in `Geometry::loadData` and projected to `apl_corr` alongside `otl_corr` in `computeSatelliteGeometry`. With the flag off, or with no usable grid, $\boldsymbol{d}_{\mathrm{APL}} = \mathbf{0}$ in Equation (1) and runs are byte-identical.

What is not modelled. Only the sun-synchronous S_1/S_2 tidal APL is modelled. The non-tidal (synoptic) pressure loading is larger and is driven by weather. Modelling it needs a surface-pressure field convolved with a Green's function, or a gridded non-tidal loading product. It is *not* modelled, and it is the leading candidate for the next site-displacement term.

1.7 Integration into the measurement model

Each epoch, the geometry builder evaluates every enabled displacement and projects it with Equation (2). The modelled-range assembly then adds the scalar corrections. In `Geometry::computeSatelliteGeometry` the OTL ECEF displacement is formed and projected to `otl_corr`, the S_1/S_2 atmospheric tide to `apl_corr`, and the solid tide is evaluated against the cached Sun/Moon positions and already returns its LOS-projected scalar. In `PppModel` the pole tide (and, when enabled, the ocean pole tide additively inside it) is projected by `computePoleTideProjection` and the base modelled value (`computeBaseModeledValue`) sums `+ otl_corr + apl_corr + solid_tide` while the pair processors add `+ pole_tide_corr`. The result is the $\rho(\boldsymbol{r})$ of the linearised range with $\boldsymbol{r}$ replaced by Equation (1).

The differenced observables of the BSSD and double-difference formulations contain whatever part of these corrections survives differencing. The solid Earth tide and the pole tide are highly correlated in space and largely cancel over short baselines. Ocean loading and atmospheric loading, the latter largely unmodelled in its non-tidal part, do not cancel. They are the dominant differential site-displacement error on long baselines.

References

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